ThaiPaliBench: Benchmarking Large Language Models on Thai-Script Pali Understanding

The first standardized benchmark for classical Pali written in Thai script — built from the Royal Pali Examination (Parian Tham) corpus, levels 1–9.

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Dataset & code
github.com/ainrg-llm-dev/ThaiPaliBench

4,000 questions, 1,000 per task

4 tasks: Grammar · Reading · NLI · Translation

9 Parian Tham proficiency levels

10 LLMs evaluated (3 commercial, 7 open-weight)

68.3% best overall accuracy (Kimi k2.5, 1T)

1 Why Thai-script Pali?

Pali is the canonical scriptural language of Theravada Buddhism and is formally taught across Thai monastic and academic institutions — **in the Thai script**. The national curriculum, **Parian Tham**, has nine graded levels (1–9) that assess grammar, translation, doctrine and composition.

THAI-SCRIPT PALI			MODERN THAI
นมตฺถุ	สมฺมา	↔ same script, different language	นอบน้อม พระสงฆ์
นโม + อตฺถุ	สมฺม + อา		pay homage (to) the Buddhist monk
sandhi	accusative		

Existing Thai benchmarks test modern, secular fluency; no exam-based resource measured whether LLMs can read Thai-script Pali. Its rich inflection, *sandhi* compounding, *scriptio continua* and "false-friend" loanwords make it a hard, low-resource case.

Contributions. (i) **ThaiPaliBench**, the first benchmark for Thai-script Pali, built from official Royal Pali Examination papers across all **9** levels; (ii) **4** discriminative tasks (MCQ + sentence-pair) enabling objective, reproducible scoring; (iii) evaluation of **10** commercial and open-weight LLMs.

2 Benchmark at a glance

Level	Grammar	RC	Transl.	NLI	Total
L1–2	500	125	125	125	875
L3	500	125	125	125	875
L4	–	125	125	125	375
L5	–	125	125	125	375
L6	–	125	125	125	375
L7	–	125	125	125	375
L8	–	125	125	125	375
L9	–	125	125	125	375
Total	**1,000**	**1,000**	**1,000**	**1,000**	**4,000**

Grammar only at foundational levels (L1–3); RC, Translation and NLI span all nine levels. Released as JSONL with source year, level, task, options and key. Licence CC BY-NC 4.0.

Grammatical Knowledge
5-option MCQ · CoLA-style

Verb conjugation, noun declension, word types, suffixes (*paccaya*), sandhi rules.

Reading Comprehension
5-option MCQ · SQuAD-style

Passage in Thai-script Pali, question answered in Thai.

Natural Language Inference
Sentence pair · XNLI-style

Pali premise + Thai hypothesis → entailment / neutral / contradiction.

Translation Understanding
5-option MCQ

Pick the most accurate Thai translation among similar-looking distractors.

2b The Parian Tham curriculum

Foundational · L1–3
Morphology, nominal suffixes, sandhi; literal vs. sense-for-sense translation; L3 adds syntactic parsing and document restructuring. Passing L3 earns the title *Maha*.

Intermediate · L4–6
More complex grammar and bidirectional Pali↔Thai translation of dense commentary prose.

Advanced · L7–9
Metrical verse composition (*Pathyāvatta*, *Vasantatilaka*) and open-ended Thai→Pali prose. L9 is recognised as equivalent to a Bachelor's degree.

Taught since the Ayutthaya period and reformed in 1912; examinations are held annually under the Sangha Supreme Council.

3 Construction pipeline

Raw PDF exams — Royal Pali Exam archives 1986–2025

VLM digitisation — Gemini 3 Pro Preview

Task synthesis — Gemini 3 Flash · MCQ + NLI pairs

ThaiPaliBench — 4,000 items · JSONL · CC BY-NC 4.0

Scanned exams (1986–2025) are digitised with a vision–language model, because conventional OCR (olmOCR-2, Typhoon OCR) made frequent errors and hallucinated on Pali diacritics. Question stems and answer keys come **directly from the official examinations**; only distractors and NLI hypotheses are LLM-generated, anchored to the official key — an **LLM-assisted** benchmark derived from official corpora.

99.7% stem provenance vs. source exam (388/389; 95% CI 98.6–100)

95.1% answer-key accuracy (370/389; 95% CI 92.5–96.9)

3,000 MCQ items run through the programmatic TRUTH ANCHOR check; 18 residual duplicate-option cases reported

< 0.15 pt estimated effect of residual artifacts on scores, per task (frozen dataset)

Manual validation on a stratified sample of 389 items (100 Grammar, 100 RC, 100 Translation, 89 NLI). Per-task key accuracy: Grammar 95.0%, RC 92.0%, Translation 94.0%, NLI 100%. Validated by the authors.

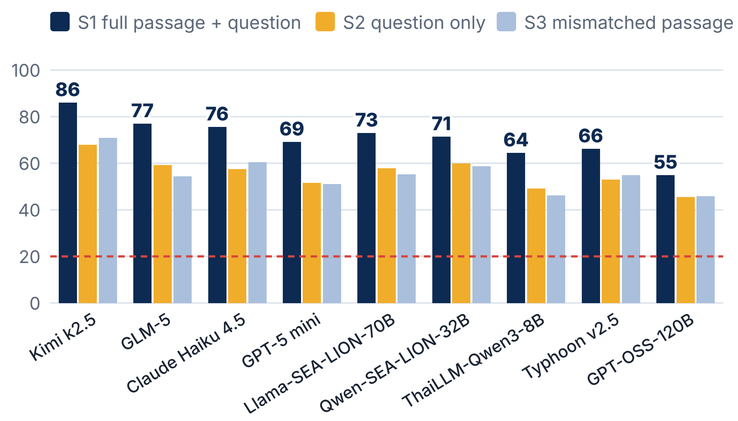
4 Evaluation setup

Commercial: GPT-5 mini, Claude Haiku 4.5, Grok 4.1 Fast.
Open-weight: Kimi k2.5 (1T), GLM-5 (744B), GPT-OSS (120B), Llama-SEA-LION-v3-70B, Qwen-SEA-LION-v4-32B, Typhoon v2.5 (30B), ThaiLLM-Qwen3-8B-Think (Pathumma).

Temperature 0.1; accuracy is the metric for every task. Gemini is excluded because it built the dataset.
Significance via **item-level paired bootstrap** (10,000 resamples) on adjacent model pairs → models are grouped into **tiers** rather than a strict ranking.

5 Does the model read, or remember?

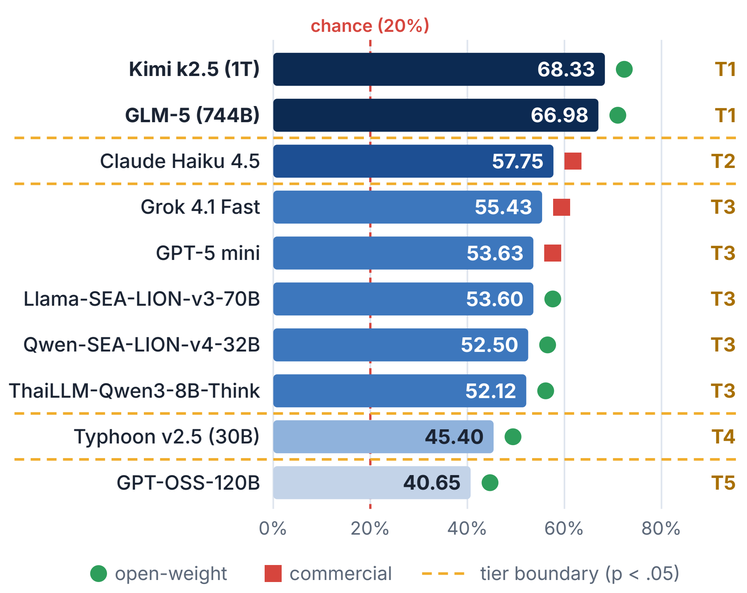
Reading comprehension stays flat across levels (69.5–73.8%). A controlled probe on the full RC subset (N = 1,000; 9 models) shows why:



RC accuracy under three context settings; numbers above bars are S1. Dashed line = 20% chance. The probe is a separate run from the main evaluation (Kimi: 86.1 here vs. 85.8); Grok 4.1 Fast was deprecated before the probe.

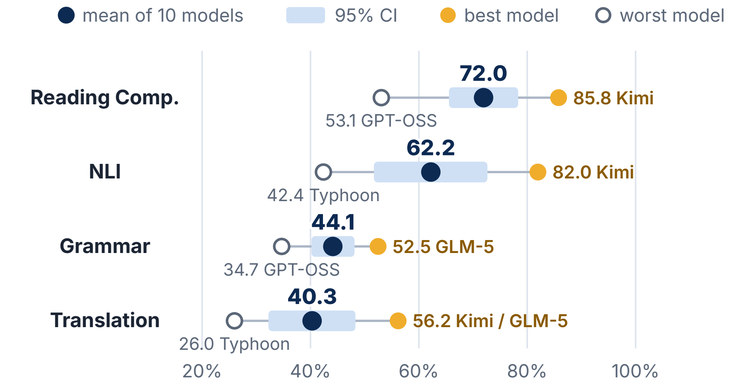
Passage reliance: removing the passage costs 9.2–18.2 points ($p < .0001$) — models do read. **Scriptural memorisation:** with no passage they still score 45.6–67.9% (2.3–3.4× chance) — canonical Buddhist texts are in pre-training data, so RC scores mix comprehension with recall.

6 Results: overall leaderboard



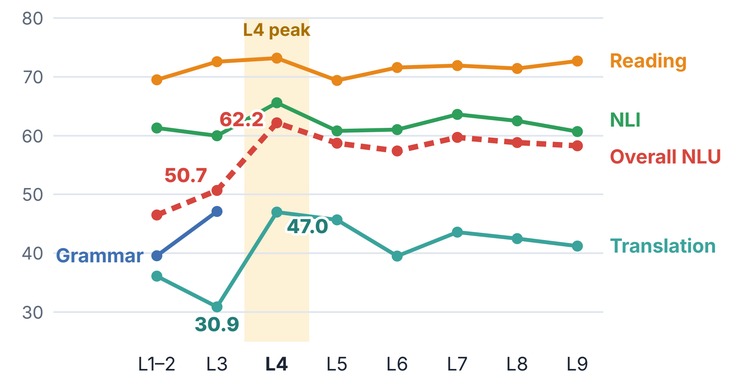
Even the best model, **Kimi k2.5**, reaches only **68.3%**; **GPT-OSS-120B** trails at **40.65%** — a 27.7-point spread. Large open-weight models (Kimi, GLM-5) **outperform every commercial model** tested. Commercial models are consistent (SD 2.1%); open-weight models vary hugely (SD 10.3%).

7 Results by task



Grammar (44.1%) and **Translation (40.3%)** are the hardest: the "Thai-to-Thai" look of the task tempts models into pattern-matching instead of real cross-lingual transfer. Open-weight models lead on Translation (42.4 vs. 35.6) and Grammar (44.6 vs. 43.0); commercial models lead on NLI (69.1 vs. 59.2) and RC (74.6 vs. 70.8).

8 Results by Parian Tham level



Category accuracy (%) by level, averaged over models. Grammar exists only at L1–3. Labelled values are from the paper text; other points are read from the paper's Fig. 5.

Difficulty is **not monotonic**: models struggle most at the "easy" foundational levels (overall NLU 50.7% at L3), peak sharply at **L4 (62.2%)** when grammar questions disappear, then plateau at 57–60% — harder exams no longer lower multiple-choice scores. Translation swings widely (30.9% → 47.0% from L3 to L4).

Key takeaways

Classical ≠ solved
Thai-script Pali remains substantially underserved: 68.3% at best, with grammar and translation below 45% on average.

Scale alone is not enough
ThaiLLM-Qwen3-8B (Thai-specialised) beats the 120B generalist GPT-OSS by 11.5 points (52.1 vs. 40.7, $p < .001$); among open-weight models, accuracy vs. size is only moderately correlated and not significant ($\rho = 0.61$, $p = 0.15$).

Reading ≠ reasoning
Models fail foundational grammar (L1–2) yet do well on advanced reading — they lean on memorised scripture rather than abstract grammatical rules.

A diagnostic resource
ThaiPaliBench surfaces gaps that modern-Thai benchmarks miss and provides a foundation for low-resource, classical-language model development.

Limitations: LLM-generated distractors and NLI hypotheses; OCR/VLM digitisation may leave minor errors; discriminative evaluation only (no generation); no certified Pali scholar reviewed the benchmark; closed APIs may drift.

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